

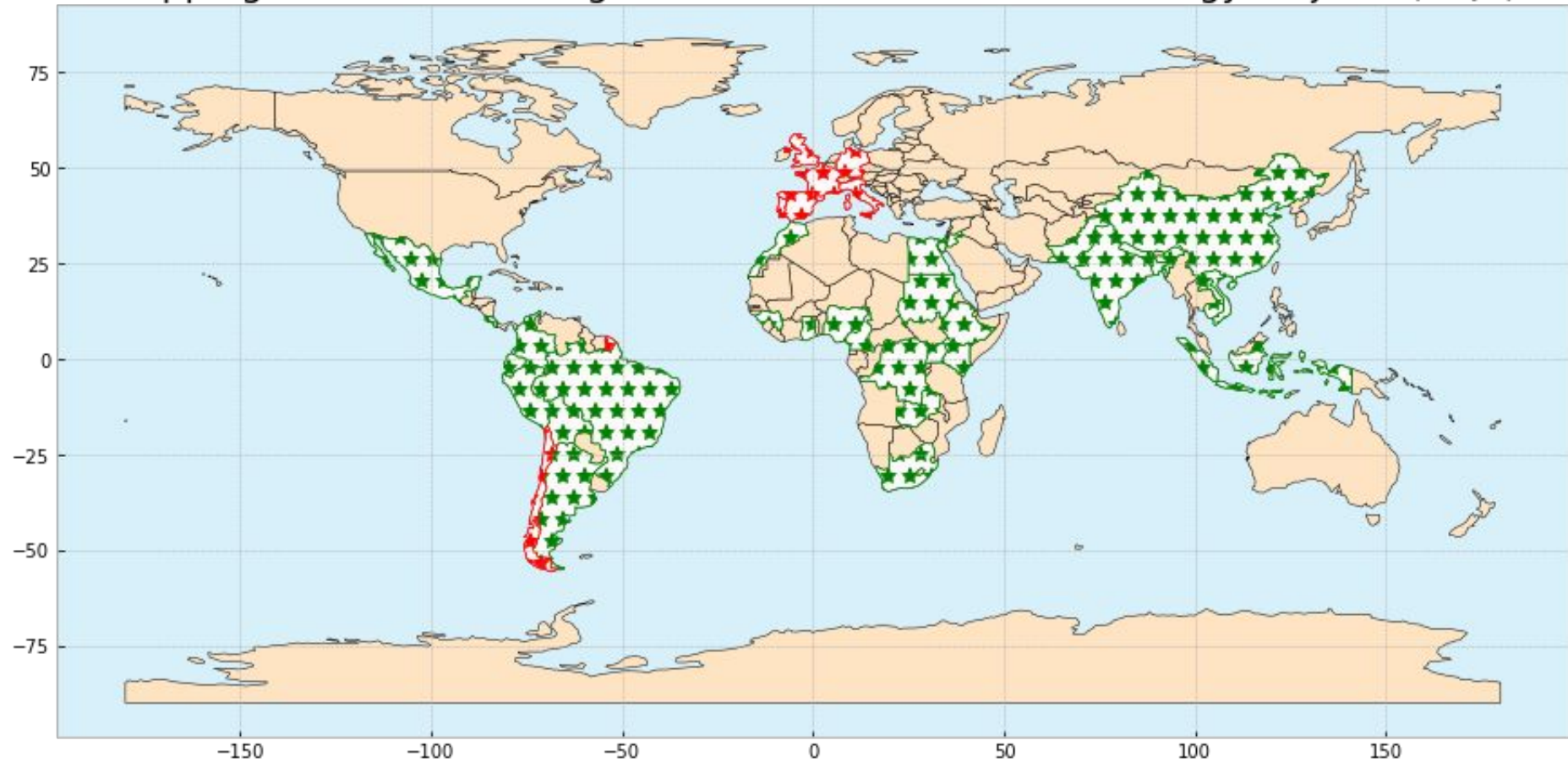
Proposing India's "The BIG PUSH" for Sustained and Sustainable Growth

Computing the Impact of Investment in Renewable Energy Projects on Output Growth of Developing Countries

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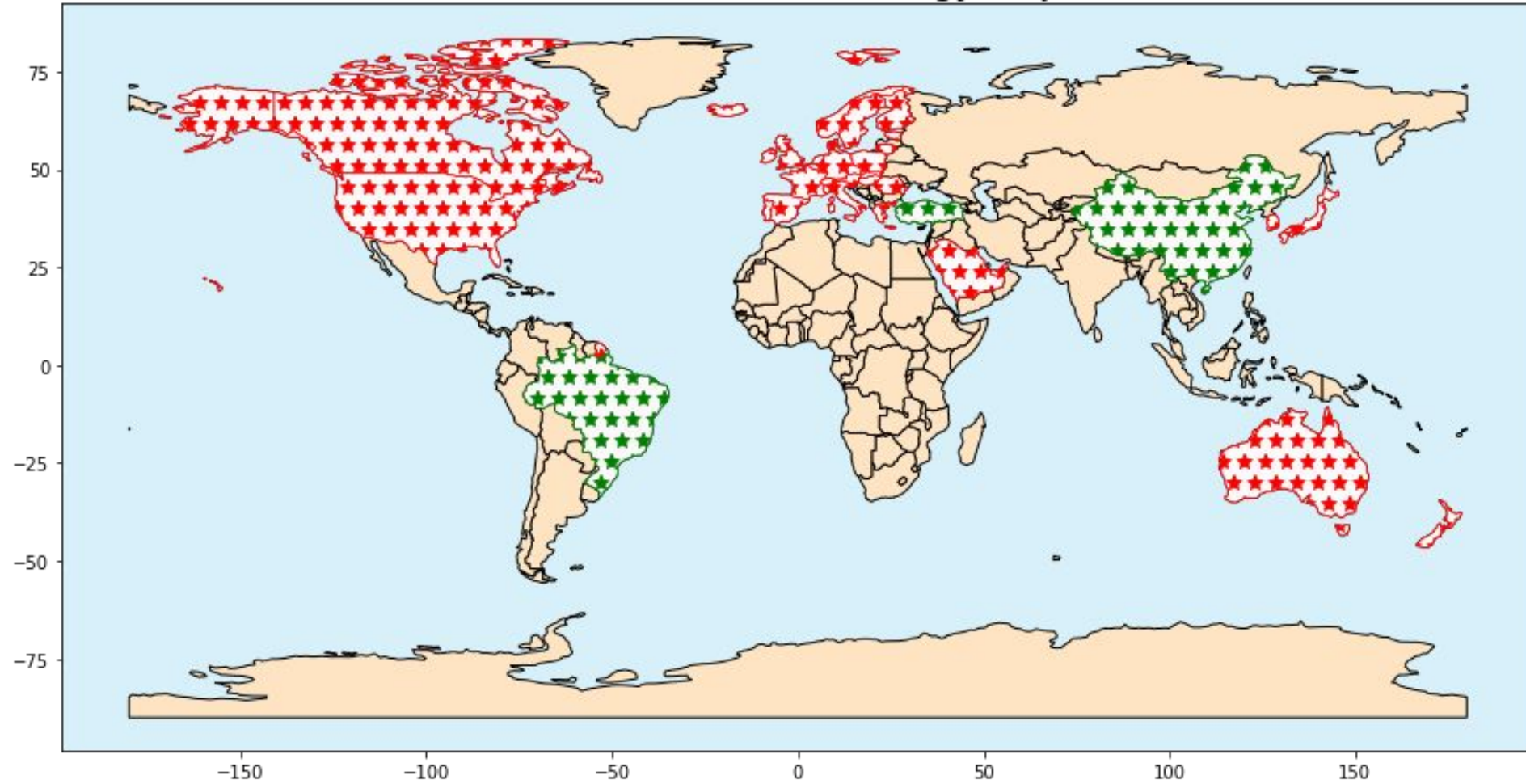
Mapping Countries with High Investment in Renewable Energy Projects (>Q3)



Agarwal,V (Source Data from International Renewable Energy Agency - IRENA)

■ Developed Countries ■ Developing Countries

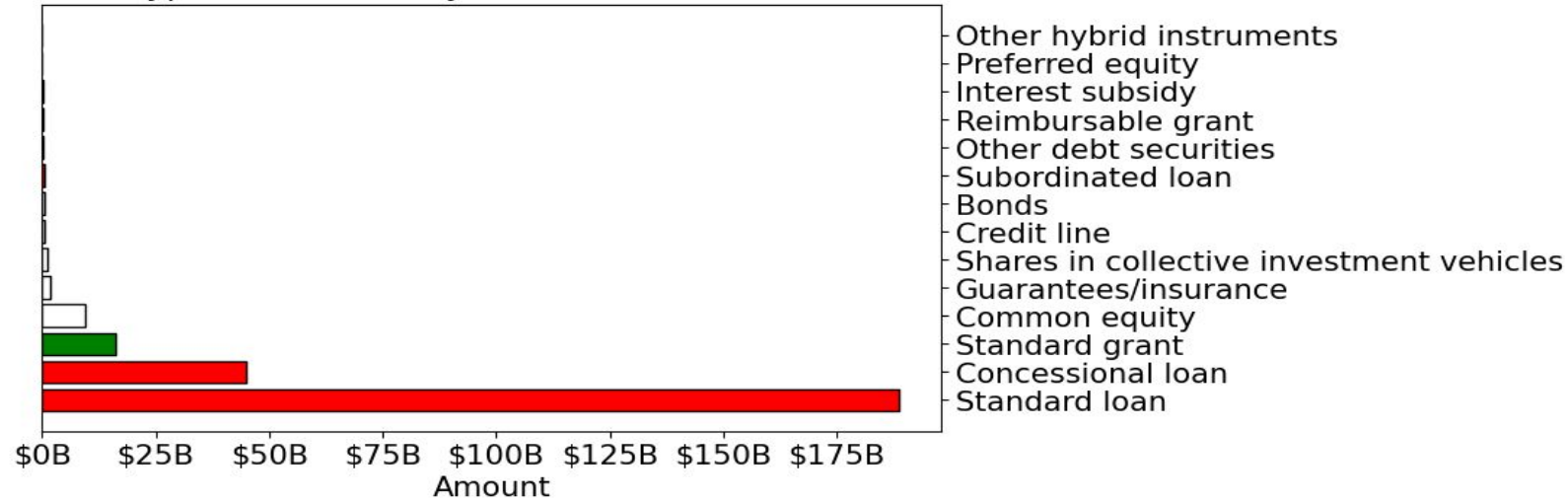
Countries with Investment in Renewable Energy Projects as Loans or Grants



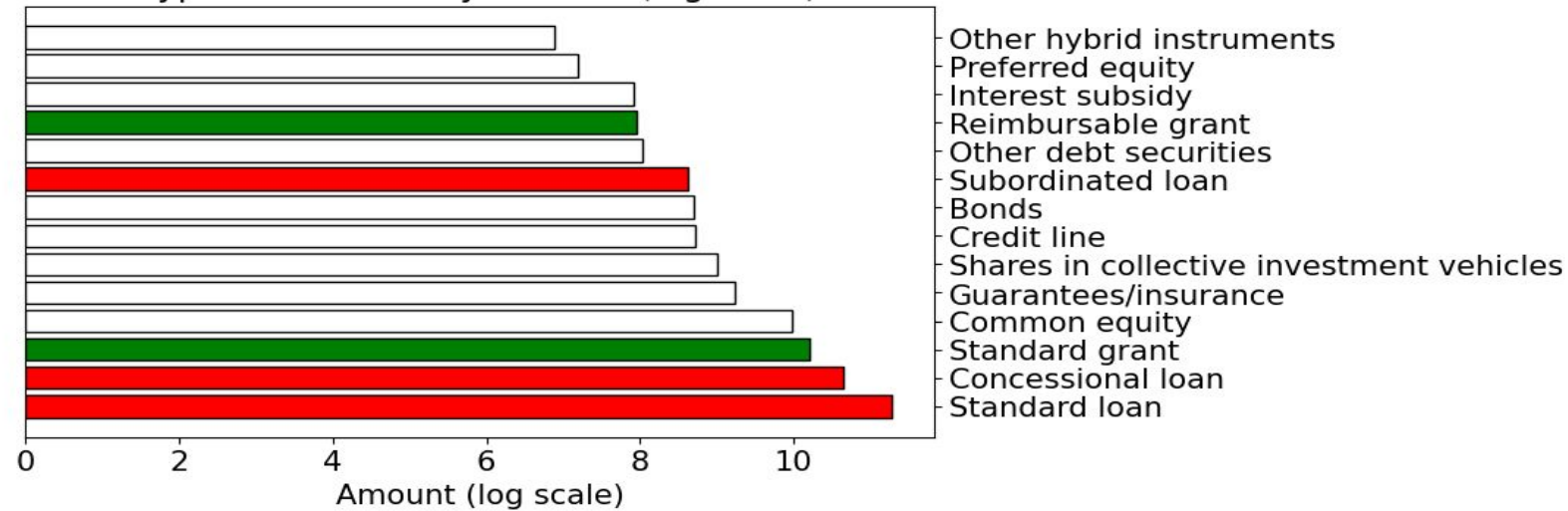
Agarwal,V (Source Data from International Renewable Energy Agency - IRENA)

Developed Countries Developing Countries

Types of Finance by Amount In USD Billion



Types of Finance by Amount (log scale)



**Will Investment in Renewable Energy Projects have
greater benefits to the Developing Countries?**



Regression Equation

$$\left(\frac{\log \frac{Y_{ijn}}{Y_{ijn0}}}{T}\right) = \beta_0 + \beta_1 \log(Y_{ijn0}) + \beta_2 \log(RE_{ij}) + \beta_3 Dev_{ij} \\ + \beta_3 \left(\frac{I}{Y}\right)_{ij} + \beta_4 \left(\frac{I}{Y}\right)_{ij}^2 + \beta_5 HCI_{ij} + \beta_6 HCI_{ij}^2 + \beta_7 (n_{ij} + \\ \gamma_{ij} + \delta_{ij}) + \beta_8 \lambda_i + \beta_9 R_j + \epsilon_{ij}$$

Where,

- $\log$ is the natural log,
- i stands for country i
- j stands for region j
- Y_{ijn} is the value of GDP in the n^{th} year
- Y_{ijn0} is the natural logarithmic of GDP in the initial year,
- β_1 is the convergence coefficient – A negative β_1 indicates convergence (developing countries are catching up with developed countries),(Chatterjee et al, 2019)
- RE stands for investment in Renewable energy projects
- Dev is a Dummy variable which takes the value 1 if the country is developing
- $(\frac{I}{Y})_{ij}$ is a proxy for physical capital accumulation taken as investment by GDP
- HCI_{ij} is a proxy for human capital accumulation using a Human Capital Index from the Penn World Table
- $(n_{ij} + \gamma_{ij} + \delta_{ij})$ is the Solow' residual taking into account changes in technology, depreciation & population growth.

Table 2. Computing the effect of Investment in Renewable Energy Projects on GDP Growth Rates with Region Fixed Effects

	<i>Dependent variable:</i>	
	Growth Rate of GDP	
	(1)	(2)
Log GDP	-0.137** (0.060)	-0.094* (0.051)
Investment/GDP	-9.772 (7.096)	-9.473 (7.115)
Secondary Schooling (%)	-2.089** (1.042)	-2.289** (1.036)
Investment(RE)	0.138 (0.090)	
Development (Developing = 1)	0.825** (0.325)	0.747** (0.314)
Secondary Schooling (%) Sq.	0.437** (0.219)	0.473** (0.218)
Investment/GDP Sq.	27.167* (15.403)	26.396* (15.443)
Solow's residual	6.540** (3.107)	6.445** (3.102)
Constant	6.513*** (1.988)	6.335*** (1.930)
Observations	80	80
R ²	0.445	0.424
Adjusted R ²	0.315	0.311
Residual Std. Error	0.678 (df = 64)	0.680 (df = 66)
F Statistic	3.422*** (df = 15; 64)	3.740*** (df = 13; 66)

Note:

*p<0.1; **p<0.05; ***p<0.01

Source: World Bank Data, Penn World Table & IRENA Renewable Energy Data

Important snippets from the Table

	With Investment in Renewable Energy Projects	Without Investment in Renewable Energy Projects
Log GDP The natural log of GDP for the initial year or β_1 is the convergence coefficient. Here a negative β_1 indicates convergence among the developing and developed countries.	-0.137^{**} (0.060)	-0.094^* (0.051)
Development (Developing = 1) Development is a dummy variable which the value 1 for developing countries. A greater magnitude indicates that countries grow at a faster rate when we account for investment in renewable energy.	0.825^{**} (0.325)	0.747^{**} (0.314)

Other Benefits of Investing in Renewable Energy



Renewable energy costs tend to be stable or decreasing over time, compared to rising or fluctuating costs for fossil fuels



Industrial and energy production are the highest contributors to rising greenhouse gas emissions.

Solution to Improving energy security, infrastructure, quality of life and employment in the rural areas of developing nations



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